

Medical Management System – Project Documentation

1. Introduction

The Medical Management System (MMS) is a web-based application designed to manage hospital operations efficiently. It integrates patient records, appointments, prescriptions, and staff management into a single platform. The system minimizes manual paperwork, improves accuracy, and enhances healthcare service delivery.

2. Objectives

- Digitize patient medical records
- Streamline appointment scheduling
- Automate billing and payment tracking
- Manage doctors and staff information
- Provide prescription and treatment history
- Ensure secure access and data privacy

3. Scope of the System

The system can be used by:

- Hospitals
- Clinics
- Diagnostic centers
- Individual practitioners

Modules included:

- Admin module

- Doctor module
- Receptionist module
- Patient module

4. System Users & Roles

4.1 Admin

- Manage users
- Configure system settings
- Manage departments

4.2 Doctor

- View appointments
- Add prescriptions
- Access patient history
- Update diagnosis

4.3 Receptionist

- Register patients
- Schedule appointments

4.4 Patient

- View appointments
- Download prescriptions
- Check billing history

5. Functional Requirements

5.1 Patient Management

- Add patient details
- Edit patient records
- Search patient
- Maintain medical history

5.2 Appointment Management

- Book appointment
- Cancel appointment
- Reschedule appointment
- Doctor availability calendar

5.3 Prescription Module

- Add medicines
- Dosage instructions
- Diagnosis notes
- Digital signature

6. System Architecture

Technologies:

- Frontend: HTML, CSS, JavaScript
- Backend: Coldfusion
- Database: MSSQL

7. Use Case Diagram (Textual Representation)

Actors:

- Admin
- Doctor
- Receptionist
- Patient

Use Cases:

- Login
- Manage Patients
- Book Appointment
- Issue Prescription

8. Data Flow Overview

Patient Registration → Appointment Booking → Consultation → Prescription → Record Storage

9. Security Measures

- Role-based access control
- Password hashing
- Session timeout
- Input validation
- SQL injection prevention
- OTP validation via email

10. Future Enhancements

- Mobile application integration
- AI-based diagnosis suggestions
- Online consultation system
- SMS/email notifications
- Insurance claim processing

11. Conclusion

The Medical Management System provides a complete digital solution for managing healthcare operations. It enhances efficiency, ensures accurate record keeping, improves patient satisfaction, and supports hospital administration with analytical insights.

12. References

- Software Engineering Principles
- Database Design Concepts
- Web Application Security Guidelines